# 1

# *Some Tricks of the Trade in Analyzing Classification*

My guess is that we have a folk theory of categorization itself. It says that things come in well-defined kinds, that the kinds are characterized by shared properties, and that there is one right taxonomy of the kinds.

It is easier to show what is wrong with a scientific theory than with a folk theory. A folk theory defines common sense itself. When the folk theory and the technical theory converge, it gets even tougher to see where that theory gets in the way—or even that it is a theory at all.

*(Lakoff 1987, 121)*

## *Introduction: A Good Infrastructure Is Hard to Find*

Information infrastructure is a tricky thing to analyze.[6] Good, usable systems disappear almost by definition. The easier they are to use, the harder they are to see. As well, most of the time, the bigger they are, the harder they are to see. Unless we are electricians or building inspectors, we rarely think about the myriad of databases, standards, and instruction manuals subtending our reading lamps, much less about the politics of the electric grid that they tap into. And so on, as many layers of technology accrue and expand over space and time. Systems of classification (and of standardization) form a juncture of social organization, moral order, and layers of technical integration. Each subsystem inherits, increasingly as it scales up, the inertia of the installed base of systems that have come before.

Infrastructures are never transparent for everyone, and their workability as they scale up becomes increasingly complex. Through due methodological attention to the architecture and use of these systems, we can achieve a deeper understanding of how it is that individuals and communities meet infrastructure. We know that this means, at the least, an understanding of infrastructure that includes these points:

- A historical process of development of many tools, arranged for a wide variety of users, and made to work in concert.
- A practical match among routines of work practice, technology, and wider scale organizational and technical resources.
- A rich set of negotiated compromises ranging from epistemology to data entry that are both available and transparent to communities of users.
- A negotiated order in which all of the above, recursively, can function together.

Table 1.1 shows a more elaborate definition of infrastructure, using Star and Ruhleder (1996), who emphasize that one person's infrastructure may be another's barrier.

This chapter offers four themes, methodological points of departure for the analysis of these complex relationships. Each theme operates as a gestalt switch—it comes in the form of an *infrastructural inversion* (Bowker 1994). This inversion is a struggle against the tendency of infrastructure to disappear (except when breaking down). It means learning to look closely at technologies and arrangements that, by design and by habit, tend to fade into the woodwork (sometimes literally!).

Infrastructural inversion means recognizing the depths of interdependence of technical networks and standards, on the one hand, and the real work of politics and knowledge production[8] on the other. It foregrounds these normally invisible Lilliputian threads and furthermore gives them causal prominence in many areas usually attributed to heroic actors, social movements, or cultural mores. The inversion is similar to the argument made by Becker (1982) in his book *Art Worlds*. Most history and social analysis of art has neglected the details of infrastructure within which communities of artistic practice emerge. Becker's inversion examines the conventions and constraints of the material artistic infrastructure and its ramifications. For example, the convention of musical concerts lasting about three hours ramifies throughout the producing organization. Parking attendants, unions, ticket takers, and theater rentals are arranged in cascading dependence on this interval of time. An eight-hour musical piece, which is occasionally written, means rearranging all of these expectations, which in turn is so expensive that such productions are rare. Or paintings are about the size, usually, that will hang comfortably on a wall. They are also the size that fits rolls of canvas, the skills of framers,

***Table 1.1***
A definition of infrastructure

- *Embeddedness.* Infrastructure is sunk into, inside of, other structures, social arrangements, and technologies,
- *Transparency.* Infrastructure is transparent to use in the sense that it does not have to be reinvented each time or assembled for each task, but invisibly supports those tasks.
- *Reach or scope.* This may be either spatial or temporal—infrastructure has reach beyond a single event or one-site practice;
- *Learned as part of membership.* The taken-for-grantedness of artifacts and organizational arrangements is a sine qua non of membership in a community of practice (Lave and Wenger 1991, Star 1996). Strangers and outsiders encounter infrastructure as a target object to be learned about. New participants acquire a naturalized familiarity with its objects as they become members.
- *Links with conventions of practice.* Infrastructure both shapes and is shaped by the conventions of a community of practice; for example, the ways that cycles of day-night work are affected by and affect electrical power rates and needs. Generations of typists have learned the QWERTY keyboard; its limitations are inherited by the computer keyboard and thence by the design of today's computer furniture (Becker 1982).
- *Embodiment of standards.* Modified by scope and often by conflicting conventions, infrastructure takes on transparency by plugging into other infrastructures and tools in a standardized fashion.
- *Built on an installed base.* Infrastructure does not grow de novo; it wrestles with the inertia of the installed base and inherits strengths and limitations from that base. Optical fibers run along old railroad lines, new systems are designed for backward compatibility; and failing to account for these constraints may be fatal or distorting to new development processes (Monteiro and Hanseth 1996).
- *Becomes visible upon breakdown.* The normally invisible quality of working infrastructure becomes visible when it breaks: the server is down, the bridge washes out, there is a power blackout. Even when there are backup mechanisms or procedures, their existence further highlights the now visible infrastructure.
- *Is fixed in modular increments, not all at once or globally.* Because infrastructure is big, layered, and complex, and because it means different things locally, it is never changed from above. Changes take time and negotiation, and adjustment with other aspects of the systems involved.[7]

Source: Star and Rohleder 1996.

and the very doorways of museums and galleries. These constraints are mutable only at great cost, and artists must always consider them before violating them.

Scientific inversions of infrastructure were the theme of a path-breaking edited volume, *The Right Tools for the Job: At Work in Twentieth-Century Life Sciences* (Clarke and Fujimura 1992). The purpose of this volume was to tell the history of biology in a new way—from the point of view of the materials that constrain and enable biological researchers. Rats, petri dishes, taxidermy, planaria, drosophila, and test tubes take center stage in this narrative. The standardization of genetic research on a few specially bred organisms (notably drosophila) has constrained the pacing of research and the ways the questions may be framed, and it has given biological supply houses an important, invisible role in research horizons. While elephants or whales might answer different kinds of biological questions, they are obviously unwieldy lab animals. While pregnant cow's urine played a critical role in the discovery and isolation of reproductive hormones, no historian of biology had thought it important to describe the task of obtaining gallons of it on a regular basis. Adele Clarke (1998) puckishly relates her discovery, found in the memoirs of a biologist, of the technique required to do so: tickle the cow's labia to make her urinate. A starkly different view of the tasks of laboratory biology emerges from this image. It must be added to the processes of stabling, feeding, impregnating, and caring for the cows involved. The supply chain, techniques, and animal handling methods had to be invented along with biology's conceptual frame; they are not accidental, but constitutive.

Our infrastructural inversion with respect to information technologies and their attendant classification systems follows this line of analysis. Like the cow's urine or the eight-hour concert, we have found many examples of counterintuitive, often humorous struggles with constraints and conventions in the crafting of classifications. For instance, as we shall see in chapter 5, in analyzing the experience of tuberculosis patients in Mann's *The Magic Mountain,* we found the story of one woman who had been incarcerated so long in the sanatorium that leaving it became unthinkable. She recovered from the disease, but tried to subvert the diagnosis of wellness. When the doctors took her temperature, she would surreptitiously dip the thermometer in hot water to make it seem that she still had a fever. On discovering this, the doctors created a thermometer without markings, so that she could not tell what the mercury column indicated. They called this

"the silent sister." The silent sister immediately becomes itself a telling indicator of the entangled infrastructure, medical politics, and the use of metrics in classifying tubercular patients. It tells a rich metaphorical story, and may become a concept useful beyond the rarified walls of the fictional Swiss asylum. What other silent sisters will we encounter in our infrastructural inversion—what surveillance, deception, caring, struggling, or negotiating?

In the sections below, four themes are presented that require the special double vision implied in the anecdotes above. They frame the new way of seeing that brings to life large-scale, bureaucratic classifications and standards. Without this map, excursions into this aspect of information infrastructure can be stiflingly boring. Many classifications appear as nothing more than lists of numbers with labels attached, buried in software menus, users' manuals, or other references. As discussed in chapter 2, new eyes are needed for reading classification systems, for restoring the deleted and dessicated narratives to these peculiar cultural, technical, and scientific artifacts.

## *Methodological Themes for Infrastructural Inversion*

### *Ubiquity*

The first major theme is the *ubiquity* of classifying and standardizing. Classification schemes and standards literally saturate our environment. In the built world we inhabit, thousands and thousands of standards are used everywhere, from setting up the plumbing in a house to assembling a car engine to transferring a file from one computer to another. Consider the canonically simple act of writing a letter longhand, putting it in an envelope, and mailing it. There are standards for paper size, the distance between lines in lined paper, envelope size, the glue on the envelope, the size of stamps, their glue, the ink in a pen, the sharpness of its nib, the composition of the paper (which in turn can be broken down to the nature of the watermark, if any; the degree of recycled material used in its production, the definition of what counts as recycling), and so forth.

Similarly, in any bureaucracy, classifications abound—consider the simple but increasingly common classifications that are used when you dial an airline for information ("if you are traveling domestically, press 1"; "if you want information about flight arrivals and departures. . . ."). And once the airline has you on the line, you are classified by them as a frequent flyer (normal, gold or platinum); corporate or

***Becoming an Irate***

Howard Becker relates a delightful anecdote concerning his classification by an airline. A relative working for one of the airlines told him how desk clerks handle customer complaints. The strategy is first to try to solve the problem. If the customer remains unsatisfied and becomes very angry in the process, the clerk dubs him or her "an irate." The clerk then calls the supervisor, "I have an irate on the line," shorthand for the category of an irritated passenger.

One day Becker was having a difficult interaction with the same airline. He called the airline desk, and in a calm tone of voice, said, "Hello, my name is Howard Becker and I'm an irate. Can you help me with this ticket?" The clerk began to sputter, "How did you know that word?" Becker had succeeded in unearthing a little of the hidden classificatory apparatus behind the scenes at the airline. He notes that the interaction after this speeded up and went particularly smoothly.

individual; tourist or business class; short haul or long haul (different fare rates and scheduling apply).

This categorical saturation furthermore forms a complex web. Although it is possible to pull out a single classification scheme or standard for reference purposes, in reality none of them stand alone. So a subproperty of ubiquity is interdependence, and frequently, integration. A systems approach might see the proliferation of both standards and classifications as purely a matter of integration—almost like a gigantic web of interoperability. Yet the sheer density of these phenomena go beyond questions of interoperability. They are layered, tangled, textured; they interact to form an ecology as well as a flat set of compatibilities. That is to say, they facilitate the coordination of heterogeneous "dispositifs techniques" (Foucault 1975). They are lodged in different communities of practice such as laboratories, records offices, insurance companies, and so forth.[9] There *are* spaces between (unclassified, nonstandard areas), of course, and these are equally important to the analysis. It seems that increasingly these spaces are marked as unclassified and nonstandard.

It is a struggle to step back from this complexity and think about the issue of ubiquity rather than try to trace the myriad connections in any one case. The ubiquity of classifications and standards is curiously difficult to see, as we are quite schooled in ignoring both, for a variety of interesting reasons. We also need concepts for under-

standing movements, textures, and shifts that will grasp patterns within the ubiquitous larger phenomenon. The distribution of residual categories ("not elsewhere classified" or "other") is one such concept. "Others" are everywhere, structuring social order. Another such concept might be what Strauss et al. (1985) call a "cumulative mess trajectory." In medicine, this occurs when one has an illness, is given a medicine to cure the illness, but incurs a serious side effect, which then needs to be treated with another medicine, and so forth. If the trajectory becomes so tangled that you cannot turn back and the interactions multiply, "cumulative mess" results. We see this phenomenon in the interaction of categories and standards all the time—ecological examples are particularly rich places to look.

***Materiality and Texture***

The second methodological departure point is that classifications and standards are *material*, as well as symbolic. How do we perceive this densely saturated classified and textured world? Under the sway of cognitive idealism, it is easy to see classifications as properties of mind and standards as ideal numbers or floating cultural inheritances. But they have material force in the world. They are built into and embedded in every feature of the built environment (and in many of the nature-culture borderlands, such as with engineered genetic organisms).

All classification and standardization schemes are a mixture of physical entities, such as paper forms, plugs, or software instructions encoded in silicon, and conventional arrangements such as speed and rhythm, dimension, and how specifications are implemented. Perhaps because of this mixture, the web of intertwined schemes can be difficult to see. In general, the trick is to question every apparently natural easiness in the world around us and look for the work involved in making it easy. Within a project or on a desktop, the seeing consists in seamlessly moving between the physical and the conventional. So when computer programmers write some lines of Java code, they move within conventional constraints and make innovations based on them; at the same time, they strike plastic keys, shift notes around on a desktop, and consult manuals for various standards and other information. If we were to try to list all the classifications and standards involved in writing a program, the list could run to pages. Classifications include types of objects, types of hardware, matches between requirements categories and code categories, and metacategories such

as the goodness of fit of the piece of code with the larger system under development. Standards range from the precise integration of the underlying hardware to the 60Hz power coming out of the wall through a standard size plug.

Merely reducing the description to the physical aspect such as the plug does not get us anywhere interesting about the actual mixture of physical and conventional or symbolic. A good operations researcher could describe how and whether things would work together, often purposefully blurring the physical and conventional boundaries in making the analysis. But what is missing is a sense of the landscape of work as experienced by those within it. It gives no sense of something as important as the texture of an organization: Is it smooth or rough? Bare or knotty? What is needed is a sense of the topography of all of the arrangements: Are they colliding, coextensive, gappy, or orthogonal? One way to get at these questions is to take quite literally the kinds of metaphors that people use when describing their experience of organizations, bureaucracies, and information systems, which are discussed in more detail in chapter 9.

When we think of classifications and standards as both material and symbolic, we adapt a set of tools not usually applied to them. There are tools for analyzing built structures, such as structural integrity, enclosures and confinements, permeability, and durability, among many others. Structures have texture and depth. The textural way of speaking of classifications and standards is common in organizations and groups. Metaphors of tautness, knots, fabrics, and networks pervade modern language (Lakoff and Johnson 1980).

### *The Indeterminacy of the Past: Multiple Times, Multiple Voices*

The third methodological theme concerns *the past as indeterminate*.[10] We are constantly revising our knowledge of the past in light of new developments in the present. This is not a new idea to historiography or to biography. We change our resumes as we acquire new skills to appear like smooth, planned paths of development, even if the change had been unexpected or undesired. When we become members of new social worlds, we often retell our life stories in new terminology. A common example of this is a religious conversion where the past is retold as exemplifying errors, sinning, and repentance (Strauss 1959). Or when one comes out as gay or lesbian, childhood behaviors and teenage crushes become indicators of early inklings of sexual choice (Wolfe and Stanley 1980).

At wider levels of scale, these revisions also mean the introduction of new voices—many possible kinds of interpretations of categories, texts, and artifacts. Multiple voices and silences are represented in any scheme that attempts to sort out the world. No one classification organizes reality for everyone—for example, the red light, yellow light, green light traffic light distinctions do not work for blind people (who need sound coding). In looking to classification schemes as ways of ordering the past, it is easy to forget those who have been overlooked in this way. Thus, the indeterminacy of the past implies recovering multivocality; it also means understanding how standard narratives that appear universal have been constructed (Star 1991a).

There is no way of ever getting access to the past except through classification systems of one sort or another—formal or informal, hierarchical or not. Take the apparently unproblematic statement: "In 1640, the English revolution occurred; this led to a twenty-year period in which the English had no monarchy." The classifications involved here, all problematic, include the following:

- The current segmentation of time into days, months, and years. Accounts of the English revolution generally use the Gregorian calendar, which was adopted some 100 years later, so causing translation problems with contemporary documents.
- The classification of peoples into English, Irish, Scots, French, and so on. These designations were by no means so clear at the time; the whole discourse of "national genius" or character only arose in the nineteenth century.
- The classification of events into revolutions, reforms, revolts, rebellions, and so forth (see Furet 1978 on thinking the French revolution). There was no concept of "revolution" at the time; our current conception is marked by the historiographical work of Karl Marx.
- What do we classify as being a "monarchy?" There is a strong historiographical tradition that says that Oliver Cromwell was a monarch—he walked, talked, and acted like one after all. Under this view, there is no hiatus at all in this English institution; rather a usurper took the throne.

There are two major historiographic schools of thought about using classification systems on the past. One maintains that we should only use classifications available to actors at the time, much as an ethnographer tries faithfully to mirror the categories of their respondents.

Authors in this tradition warn against the dangers of anachronism. Hacking (1995) on child abuse is a sophisticated version that we discuss in chapter 7. If a category did not exist contemporaneously, it should not be retroactively applied.

The other school of thought holds that we should use the real classifications that progress in the arts and sciences has uncovered. Often history informed by current sociology will take this path. For example, Tort's (1989) work on "genetic" classification systems (which were not so called at the time, but which are of vital interest to the Foucaldian problematic) imposes a post hoc order on nineteenth-century classification schemes in a variety of sciences. Even though those schemes were perceived by their creators as responding solely to the specific needs of the discipline they were dealing with (etymology, say, or mineralogy), Tort demonstrates that there was a link between many different schemes (both direct in people shifting disciplines and conceptual in their organization) that allows us to perceive an order nowhere apparent to contemporaries.

From a pragmatist point of view, both aspects are important in analyzing the consequences of modern systems of classification and standardization. We seek to understand classification systems according to the work that they are doing and the networks within which they are embedded. That entails both an understanding of the categories of those designing and using the systems, and a set of analytic questions derived from our own concerns as analysts.

When we ask historical questions about the deeply and heterogeneously structured space of classification systems and standards, we are dealing with a four-dimensional archaeology. The systems move in space, time, and process. Some of the archaeological structures we uncover are stable, some in motion, some evolving, some decaying. They are not consistent. An institutional memory about an epidemic, for example, can be held simultaneously and with internal contradictions (sometimes piecemeal or distributed and sometimes with entirely different stories at different locations) across a given institutional space.

In the case of AIDS, classifications have shifted significantly over the last twenty years, including the invention of the category in the 1980s—from gay-related immune disorder (GRID) through a chain of other monikers to the now accepted acquired immune deficiency syndrome (AIDS). It is now to some extent possible to look back at cases that might previously have been AIDS (Grmek 1990) before we had

***When Is It a Harley?***

One of the ways the past becomes indeterminate is through gradual shifts in what it means to "really be" something—the essence of it.

Sitting in a tattoo parlor, surrounded by people I do not usually hang out with. Young men in black leather vests and sun-bleached hair. I turn to the waiting room reading material, which in this case is the monthly *Thunder Press,* a newsletter for motorbike aficionados. The lead article asks the question: "Is It Still a Harley" if you have customized your bike yourself? The Oregon Department of Motor Vehicles makes the definitive call: "Anything that is not totally factory built will make it a reconstructed motorcycle, and it will be called 'assembled' on the title" (69).

A major activity in the Harley social world is customizing features of one's motorcycle, and there are important symbolic and affiliative signs attached to the customizing process. Deleting the name Harley from the registration form is perceived as an insult to the owner, and this insult is stitched together in the article with others that come from the government toward bikers (restricting meeting places, insisting on helmet-wearing, being overly enthusiastic in enforcing traffic violations by bikers).

This is a pure example of the politics of essence, of identity politics. It is echoed in many areas of life, for example, in James Davis' (1991) classic study *Who Is Black?* where the question of the one-drop rule in the United States, and the rejection of mixed-race people as a legitimate category is an old and a cruel story. The central process here is the distillation of the sine qua non out from the messy and crenellated surrounds—the rejection of marginality in favor of purity.

When this occurs, the suffering of the marginal becomes privatized and distributed, creating the conditions for pluralistic ignorance ("I'm the only one"). Meeting the purity criteria of the essentialized category also becomes bureaucratized and again the onus is shifted to the individual alone. Only when the category is joined with a social movement can the black box of essence be reopened, as for example with the recent uprisings and demonstrations of mixed race Hispanic people toward the U.S. census and its rigid categories. The problem becomes clear if one is both black and Hispanic, a common combination in the Caribbean. Through which master trait will the government perceive you?

—Leigh Star

Source: Anonymous, "Is It Still a Harley," *Thunder Press* 5:4 (July 1996, 1 and 69).

the category (a problematic gaze to be sure, as Bruno Latour (forthcoming) has written about tuberculosis). There are epidemiological stories about trying to collect information about a shameful disease; there is a wealth of personal and public narratives about living with it. There is a public health story and a virology story, which use different category systems. There are the standardized forms of insurance companies and the categories and standards of the Census Bureau. When an attempt was made to combine these data in the 1980s to disenfranchise young men living in San Francisco, from health insurance, the resultant political challenge stopped the combination of these data from being so used. At the same time, the San Francisco blood banks refused for years to employ HIV screening, thus denying the admission of another category to their blood labeling, as Shilts (1987) tells us, with many casualties as a result. Whose story has categorical ascendancy here? That question is forever morally moot—all of the stories are important and all of the categories tell a different one.

***Practical Politics***

The fourth major theme is uncovering *the practical politics of classifying and standardizing*. This is the design end of the spectrum of investigating categories and standards as technologies. There are two processes associated with these politics: arriving at categories and standards, and, along the way, deciding what will be visible or invisible within the system.

It follows from the indeterminacy discussed above that the spread or enforcement of categories and standards involves negotiation or force. Whatever appears as universal or indeed standard, is the result of negotiations, organizational processes, and conflict. How do these negotiations take place? Who determines the final outcome in preparing a formal classification? Visibility issues arise as one decides where to make cuts in the system, for example, down to what level of detail one specifies a description of work, of an illness, of a setting. Because there are always advantages and disadvantages to being visible, this becomes crucial in the workability of the schema. As well, ordinary biases of what should be visible, or legitimated, within a particular scheme are always in action. The trade-offs involved in this sort of politics are discussed in chapters 5 on tuberculosis and 7 on nursing work.

Someone, somewhere, must decide and argue over the minutiae of classifying and standardizing. The negotiations themselves form the

***There's No Such Thing as a Rodent***

An article in the *San Jose Mercury News* by Rick Weiss declares: "Researchers say there's no such thing as a rodent." He quotes an article from *Nature,* which argues that the 2,000 species of animals ordinarily considered rodents—including rats, mice, and guinea pigs—did not evolve from a common ancestor. The finding is deeply controversial. Weiss says, "On one side are researchers who have spent their careers hunched over fossils or skeletal remains to determine which animals evolved from which." On the other, the article continues, are those who would use DNA analysis to make the determination. The fossil studiers say that DNA is not yet accurate enough. The classification of species has always been deeply controversial. Biologists speak of a rough cut among their ranks: lumpers (those who see fewer categories and more commonalties) versus splitters (those who would name a new species with fewer kinds of difference cited). There are always practical consequences for these names. Splitters, for example, often included people who wanted a new species named after them, and the more species there are, the more likely is an eponymous label. The deliberately provocative headline of this article demands a response: "well, don't tell that to my cat." We often refer implicitly in this fashion to the power of naming—blurring the name of the category with its members. (*San Jose Mercury News,* June 13, 1996: 5A by Rick Weiss)

basis for a fascinating practical ontology—our favorite example is when is someone really alive? Is it breathing, attempts at breathing, or movement? And how long must each of those last? Whose voice will determine the outcome is sometimes an exercise of pure power: We, the holders of western medicine and scions of colonial regimes, will decide what a disease is and simply obviate systems such as acupuncture or Aryuvedic medicine. Sometimes the negotiations are more subtle, involving questions such as the disparate viewpoints of an immunologist and a surgeon, or a public health official (interested in even *one* case of the plague) and a statistician (for whom one case is not relevant).

Once a system is in place, the practical politics of these decisions are often forgotten, literally buried in archives (when records are kept at all) or built into software or the sizes and compositions of things. In addition to our archaeological expeditions into the records of such negotiations, this book provides some observations of the negotiations in action.

Finally, even where everyone agrees on how classifications or standards should be established, there are often practical difficulties about how to craft them. For example, a classification system with 20,000 bins on every form is practically unusable for data-entry purposes. The constraints of technological record keeping come into play at every turn. For example, the original ICD had some 200 diseases not because of the nature of the human body and its problems but because this was the maximum number that would fit the large census sheets then in use.

Sometimes the decision simply about how fine-grained to make the system has political consequences as well. For instance, describing and recording someone's tasks, as in the case of nursing work, may mean controlling or surveilling their work as well, and may imply an attempt to take away discretion. After all, the loosest classification of work is accorded to those with the most power and discretion who are able to set their own terms. There are financial stakes as well. In a study of a health insurance company's system of classifying for doctor and patient reimbursement, Gerson and Star (1986) found that doctors wanted the most fine-grained of category systems, so that each procedure could be reimbursed separately and thus most profitably. Data-entry personnel and hospital administrators, among others, wanted broader, simpler, and coarser-grained categories for reasons of efficiency. These conflicts were, however, invisible to the outside world, which received only the forms for reimbursement purposes and a copy of the codebook for reference. Both the content of the categories and the structure of the overall scheme are concerns for due process within organizations—whose voice will be heard and when will enough data, of the right granularity, have been collected?

### *Infrastructure and Method: Convergence*

These ubiquitous, textured classifications and standards help frame our representation of the past and the sequencing of events in the present. They can best be understood as doing the ever local, ever partial work of making it appear that science describes nature (and nature alone) and that politics is about social power (and social power alone). Consider the case of psychoanalysts discussed at length in Young (1995), Kirk and Kutchins (1992), and Kutchins and Kirk (1997). To receive reimbursement for their procedures, psychoanalysts now need to couch them in a biomedical language (using the DSM).

***Fitting Categories to Circumstances***

An academic friend on the East Coast tells an anecdote of negotiation with her long-term psychoanalyst about how to fill out her insurance forms. She was able to receive several free sessions of therapy a year under her health insurance plan. Each year, she and her therapist would discuss how best to categorize her. It was important to represent the illness as serious and long-term. At the same time, they were worried that the information about the diagnosis might not always remain confidential. What could they label her that would be both serious and nonstigmatizing? Finally, they settled on the diagnosis of obsessive-compulsive. No academic would ever be penalized for being obsessive-compulsive, our friend concluded with a wry laugh! (Kirk and Kutchins (1992) document similar negotiations between psychiatrists and patients.)

Theoretically, this rubric is anathema to them, systematically replacing the categories of psychoanalysis with the language of the pharmacopoeia and of the biochemistry of the brain. The DSM, however, is the lingua franca of the medical insurance companies. Thus, psychoanalysts use the categories not only to obtain reimbursement but as a shorthand to communicate with each other. There are local translation mechanisms that allow the DSM to continue to operate in this fashion and, at the same time, to become the sole legal, recognized representation of mental disorder. A "reverse engineering" of the DSM or the ICD reveals the multitude of local political and social struggles and compromises that go into the constitution of a "universal" classification.

Standards, categories, technologies, and phenomenology are increasingly converging in large-scale information infrastructure. As we have indicated in this chapter, this convergence poses both political and ethical questions. These questions are by no means obvious in ordinary moral discourse. For all the reasons given above, large-scale classification systems are often invisible, erased by their naturalization into the routines of life. Conflict and multiplicity are often buried beneath layers of obscure representation.

Methodologically, we do not stand outside these systems, nor pronounce on their mapping to some otherworldly "real" or "constructed" nature. Rather, we are concerned with what they do, pragmatically speaking, as scaffolding in the conduct of modern life. Part of that

analysis means understanding the coconstruction of classification systems with the means for data collection and validation.

To clarify our position here, let us take an analogy. In the early nineteenth century in England there were a huge number of capital crimes, starting from stealing a loaf of bread and going on up. Precisely because the penalties were so draconian, however, few juries would ever impose the maximum sentence; and indeed there was a drastic reduction in the number of executions even as the penal code was progressively strengthened. There are two ways of writing this history: one can either concentrate on the creation of the law; or one can concentrate on the way things worked out in practice. This is very similar to the position taken in Latour's *We Have Never Been Modern* (1993). He argues that we can either look at what scientists say they are doing (working within a purified realm of knowledge) or at what they actually are doing (manufacturing hybrids of nature-culture). We think both are important. We advocate here a pragmatic methodological development—pay more attention to the classification and standardization work that allows for hybrids to be manufactured and so more deeply explore the terrain of the politics of science in action.

The point is that both words and deeds are valid kinds of account. Early sociology of science in the actor-network tradition concentrated on the ways in which it comes to appear that science gives an objective account of natural order: trials of strength, enrolling of allies, cascades of inscriptions, and the operation of immutable mobiles (Latour 1987, 1988). Actor network theory drew attention to the importance of the development of standards (though not to the linked development of classification systems), but did not look at these in detail. Sociologists of science invited us to look at the process of producing something that looked like what the positivists alleged science to be. We got to see the Janus face of science as both constructed and realist. In so doing we followed the actors, often ethnographically. We shared their insights. Allies must be enrolled, translation mechanisms must be set in train so that, in the canonical case, Pasteur's laboratory work can be seen as a direct translation of the quest for French honor after defeat in the battlefield (Latour 1988).

By the very nature of the method, However, we also shared the actors' blindness. The actors being followed did not themselves *see* what was excluded: they constructed a world in which that exclusion could occur. Thus if we just follow the doctors who create the ICD at the WHO in Geneva, we will not see the variety of representation

systems that other cultures have for classifying diseases of the body and spirit; and we will not see the fragile networks these classification systems subtend. Rather, we will see only those who are strong enough and shaped in such a fashion as to impact allopathic medicine. We will see the blind leading the blind.

This blindness occurs by changing the world such that the system's description of reality becomes true. Thus, for example, consider the case where all diseases are classified purely physiologically. Systems of medical observation and treatment are set up such that physical manifestations are the only manifestations recorded. Physical treatments are the only treatments available. Under these conditions, then, logically schizophrenia may only result purely and simply from a chemical imbalance in the brain. It will be impossible to think or act otherwise. We have called this the principle of *convergence* (Star, Bowker and Neumann in press).

### *Resistance*

Reality is 'that which resists,' according to Latour's (1987) Pragmatist-inspired definition. The resistances that designers and users encounter will change the ubiquitous networks of classifications and standards. Although convergence may appear at times to create an inescapable cycle of feedback and verification, the very multiplicity of people, things and processes involved mean that they are never locked in for all time.

The methods in this chapter offer an approach to resistance as a reading of where and how political work is done in the world of classifications and standards, and how such artifacts can be problematized and challenged. Donald MacKenzie's (1990) wonderful study of "missile accuracy" furnishes the best example of this approach. In a concluding chapter to his book, he discusses the possibility of "uninventing the bomb," by which he means changing society and technology in such a way that the atomic bomb becomes an impossibility. Such change, he suggests, can be carried out in part at the overt level of political organizations. Crucially for our purposes, however, he also sensitizes the reader to the site of the development and maintenance of technical standards as a site of political decisions and struggle. Standards and classifications, however dry and formal on the surfaces, are suffused with traces of political and social work. Whether we wish to uninvent any particular aspect of complex information infra-

structure is properly a political and a public issue. Because it has rarely been cast in that light, tyrannies of various sorts flourish. Some are the tyrannies of inertia—red tape—rather than explicit public policies. Others are the quiet victories of infrastructure builders inscribing their politics into the systems. Still other are almost accidental—systems that become so complex that no one person and no organization can predict or administer good policy.

The magic of modern technoscience is a lot of hard work involving smoke-filled rooms, and boring lists of numbers and settings. Tyranny or democracy, its import on our lives cannot be denied. This chapter has offered a number of points of departure for evaluation, resistance, and better analysis of one of its least understood aspects.